In [2]:
```
import pandas as pd
import matplotlib.pyplot as plt

# 1. Read dataset
df = pd.read_csv("Retail_Sales_Data.csv")

# 2. Explore dataset
print(df.head())
print(df.info())
print(df.describe())

# 3 & 4. Total sales by product category
sales = df.groupby("Product Category")["Total Amount"].sum()

print("\nSales by Product Category:")
print(sales)

# 5. Bar plot
sales.plot(kind="bar")
plt.xlabel("Product Category")
plt.ylabel("Total Sales")
plt.title("Sales by Product Category")
plt.show()

# 6. Top-performing category
print("\nTop Performing Category:")
print(sales.idxmax())
print("Sales:", sales.max())

# 7. Sales by Gender and Product Category
category_sales = df.groupby(
    ["Gender", "Product Category"]
)["Total Amount"].sum()

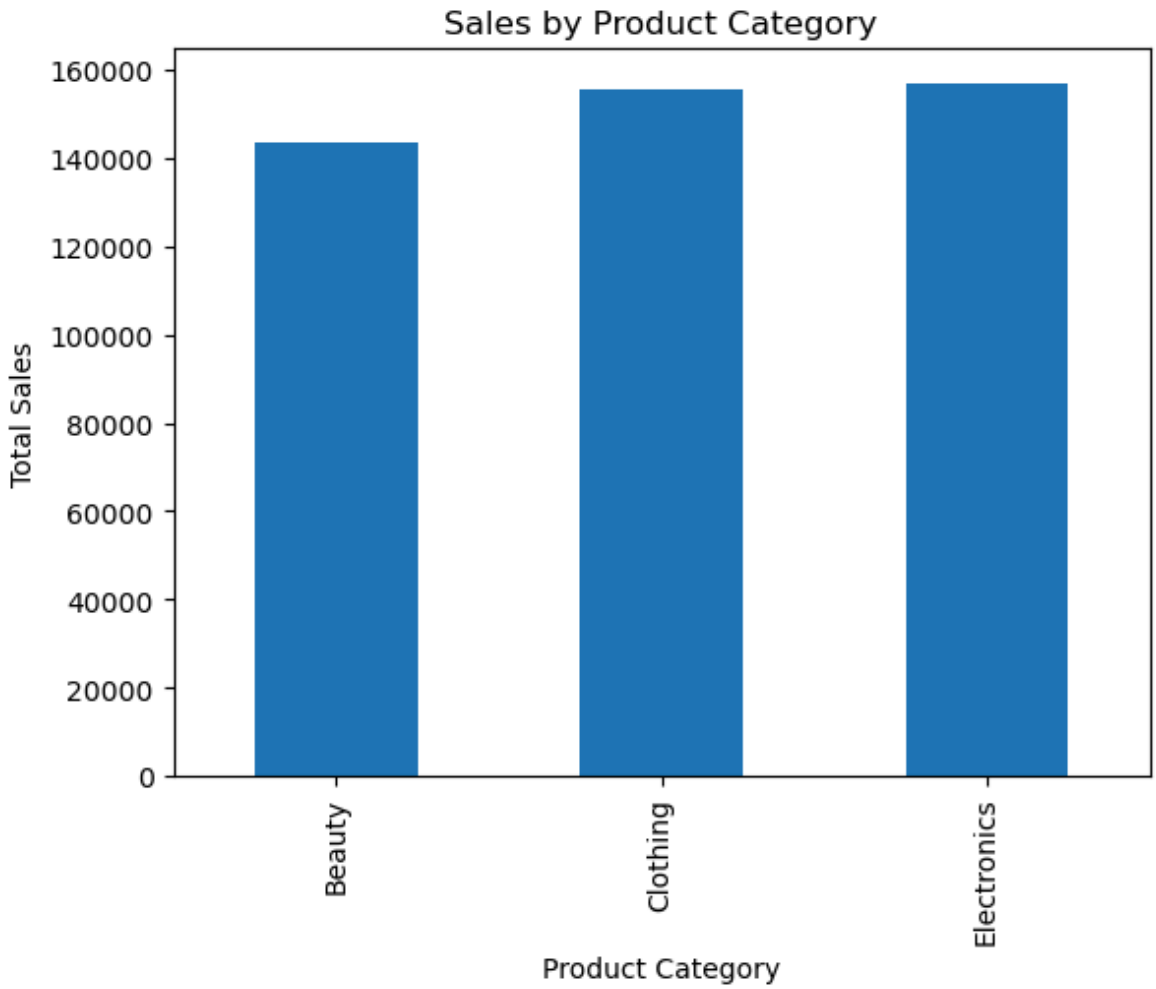
print("\nSales by Gender and Product Category:")
print(category_sales)

# 8. Grouped bar plot
category_sales.unstack().plot(kind="bar")
plt.xlabel("Gender")
plt.ylabel("Total Sales")
plt.title("Sales by Gender and Product Category")
plt.show()
```

```
   Transaction ID        Date Customer ID  Gender  Age Product Category  \
0               1  2023-11-24     CUST001    Male   34           Beauty
1               2  2023-02-27     CUST002  Female   26         Clothing
2               3  2023-01-13     CUST003    Male   50      Electronics
3               4  2023-05-21     CUST004    Male   37         Clothing
4               5  2023-05-06     CUST005    Male   30           Beauty

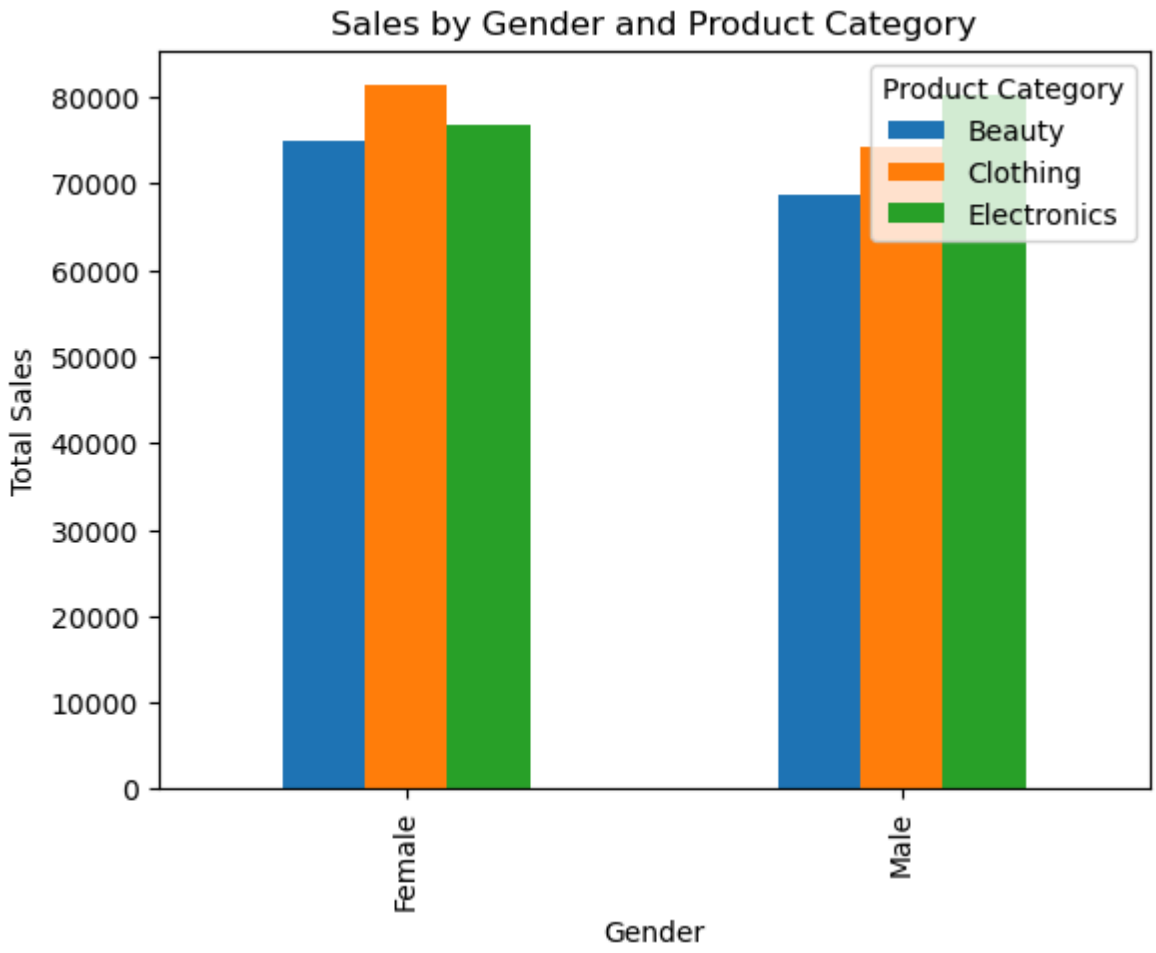
   Quantity  Price per Unit  Total Amount
0         3              50           150
1         2             500          1000
2         1              30            30
3         1             500           500
4         2              50           100
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 9 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   Transaction ID    1000 non-null   int64
 1   Date              1000 non-null   object
 2   Customer ID       1000 non-null   object
 3   Gender            1000 non-null   object
 4   Age               1000 non-null   int64
 5   Product Category  1000 non-null   object
 6   Quantity          1000 non-null   int64
 7   Price per Unit    1000 non-null   int64
 8   Total Amount      1000 non-null   int64
dtypes: int64(5), object(4)
memory usage: 70.4+ KB
None
       Transaction ID         Age     Quantity  Price per Unit  Total Amount
count     1000.000000  1000.00000  1000.000000     1000.000000   1000.000000
mean       500.500000    41.39200     2.514000      179.890000    456.000000
std        288.819436    13.68143     1.132734      189.681356    559.997632
min          1.000000    18.00000     1.000000       25.000000     25.000000
25%        250.750000    29.00000     1.000000       30.000000     60.000000
50%        500.500000    42.00000     3.000000       50.000000    135.000000
75%        750.250000    53.00000     4.000000      300.000000    900.000000
max       1000.000000    64.00000     4.000000      500.000000   2000.000000

Sales by Product Category:
Product Category
Beauty         143515
Clothing       155580
Electronics    156905
Name: Total Amount, dtype: int64
```




```
Top Performing Category:
Electronics
Sales: 156905

Sales by Gender and Product Category:
Gender  Product Category
Female  Beauty              74830
        Clothing            81275
        Electronics         76735
Male    Beauty              68685
        Clothing            74305
        Electronics         80170
Name: Total Amount, dtype: int64
```


Sales by Gender and Product Category
Product Category
Beauty
Clothing
Electronics
Total Sales
80000
70000
60000
50000
40000
30000
20000
10000
0
Female
Male
Gender

In [ ]: